Assignment 7: Retail Billing

Course: CS101 Computational Thinking | B.Tech 1st Year

Problem Statement

Process a customer's shopping cart consisting of multiple products. Calculate taxes based on categories, aggregate the total bill, and apply a cart-level threshold discount while enforcing a maximum concession limit.

Algorithm

1. Create a list for the cart items (Item, Quantity, Price, Category).
2. Initialize a variable for the grand total.
3. Iterate through items using a `for` loop. Calculate base cost (Qty * Price).
4. Apply tax rates based on the category string via conditionals (e.g., Electronics 18%, Grocery 0%).
5. Accumulate the post-tax cost of each item.
6. Apply global discounts to the final sum, capping the discount at a predefined maximum limit.

Python Implementation

```
# (Product, Quantity, Unit_Price, Category)
cart = [
    ("Smartphone", 1, 15000, "Electronics"),
    ("Jeans", 2, 1200, "Apparel"),
    ("Rice", 1, 500, "Grocery")
]

subtotal = 0.0
print("--- Itemized Invoice ---")

for item in cart:
    name, qty, price, category = item
    item_cost = qty * price

    # Category-specific tax
    tax = 0.0
    if category == "Electronics":
        tax = item_cost * 0.18
    elif category == "Apparel":
        tax = item_cost * 0.05

    final_item_price = item_cost + tax
    subtotal += final_item_price
    print(name + " x" + str(qty) + " : Rs. " + str(final_item_price))

# Threshold logic and Maximum Concession Limit
discount = 0.0
if subtotal > 10000:
    discount = subtotal * 0.10 # 10% off

# Enforce Maximum Concession Limit of Rs. 1500
if discount > 1500:
    discount = 1500

grand_total = subtotal - discount

print("------------------------")
```